Stationary Power-Law Solutions of Weak Kinetic-Alfvénic Turbulence

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Kinetic Alfvén Wave

- Kinetic Alfvén wave (KAW) with FILR correction exists in
  - solar wind, solar corona, interstellar medium, magnetically confined fusion
- Coherent mode-coupling processes (gyrokinetic framework)
  - parametric decay: KAW $\longrightarrow$ KAW + ion sound wave (ISW) [L.Chen & F.Zonca EPL 2011]
  - modulational instability: KAW $\longrightarrow$ KAW + convective cell (CC) [F.Zonca et al. EPL 2015]
  - parametric decay: KAW $\longrightarrow$ KAW + KAW [K.Shen et al. PoP 2024]

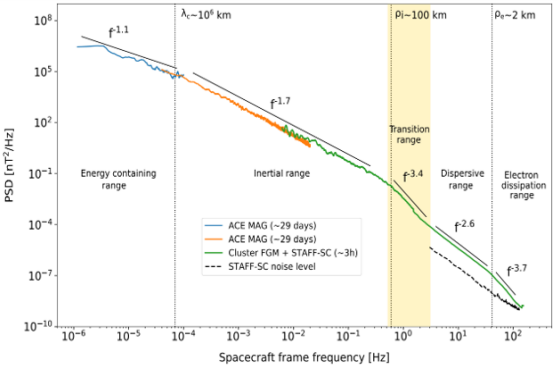
$\Longrightarrow$ Low-frequency kinetic-Alfvénic turbulence (three-KAW interactions)

- burning plasmas: heating, turbulent saturation, anomalous transport
- space plasmas: stationary spectra, particle heating and acceleration

Solar Wind

Strong KAW Turbulence

- Stationary spectra: multiscale and Alfvénic
  - ideal MHD range $E(k_\perp) \propto k_\perp^{-1.7}$
  - ion-kinetic/dispersive range $k_\perp^{-2.6}$
- Imbalanced solar wind: unequal fluxes of Alfvénic fluctuations propagating in each direction along the mean field
  - transition range $(k_\perp \rho_i \lesssim 1)$: $k_\perp^{-4} \sim k_\perp^{-3}$
  - non-zero helicity $\Longrightarrow$ inverse cascade identified numerically by [J.Cho PRL 2011; G.Miloshevich et al. JPP 2021]

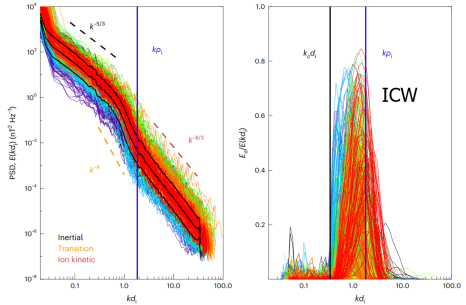


Taylor's hypothesis: $f \Leftrightarrow k_\perp$
[F.Sahraoui RMPP 2020]

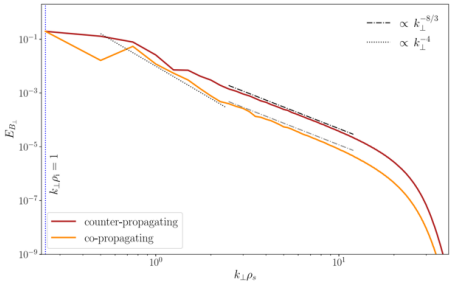
Solar Wind

Strong KAW Turbulence

- Two possible mechanisms for the formation of transition range:
  - "helicity barrier" due to conservation of helicity at sub-ion scales [R.Meyrand et al. JPP 2021]
  - steep spectrum due to nonlinear interactions among co-propagating KAWs [T.Passot et al. JPP 2022]



Excitation of ICWs due to the helicity barrier[T.A.Bowen et al. Nat.Astron. 2024]



Gyrofluid simulation of imbalanced KAW turbulence[T.Passot et al. FPP 2024]

Weak Turbulence

- Motivation: the role of helicity and nonlinear interactions among co-propagating waves in weak KAW turbulence
- Weak KAW turbulence $\Longrightarrow$ wave-kinetic description [R.Z.Sagdeev & A.A.Galeev 1969]
  - weak-nonlinearity & random phase approximation
- Stationary spectra $\Longrightarrow$ Kolmogorov-Zakharov (KZ) spectra [V.E.Zakharov et al. 1992]
  - a set of power-law solutions after transforming the wave kinetic equation
  - energy-cascade spectrum of reduced-MHD turbulence: $E(k_\perp) \propto k_\perp^{-2}$ [S.Boldyrev & J.C.Perez PRL 2009]
  - energy-cascade spectrum of electron-MHD turbulence: $E(k_\perp) \propto k_\perp^{-5/2}$ [V.David et al. PRL 2024]
- We investigate KZ spectra corresponding to energy-cascade and helicity-cascade in both $k_\perp \rho_i \ll 1$ and $k_\perp \rho_i \gg 1$ limits, for both co-propagating and counter-propagating cases.

Gyrokinetic Model Equations

- Assumptions: $\omega \ll \Omega_{ci}$, $k_\parallel \ll k_\perp$, $1 \gg \beta_i \sim \beta_e \gg m_e/m_i$, uniform system
- Generalized parallel Ohm's law

$$c^{-1}\partial_t \delta A_\parallel + \nabla_\parallel \sigma_0 \delta\phi = B_0^{-1}[\delta A_\parallel, \sigma_0 \delta\phi] \tag{2.1}$$

- Generalized nonlinear gyrokinetic vorticity equation

$$(\Gamma_0 - 1)\partial_t \delta\phi + \frac{\rho_i^2 v_A^2}{c}\nabla_\parallel \Delta_\perp \delta A_\parallel = -\frac{c}{B_0}[\delta\phi, \Gamma_0 \delta\phi] + \frac{\rho_i^2 v_A^2}{cB_0}[\delta A_\parallel, \Delta_\perp \delta A_\parallel] \tag{2.2}$$

FILR effect fully retained

- Generalized free-energy (wave energy) and generalized helicity (parallel-momentum)
  [A.A.Schekochihin et al. ApJS 2009; R.Meyrand et al. JPP 2021]

$$E = \int d\mathbf{r}\left[\frac{n_0 e^2}{T_i}\frac{\sigma_0(1-\Gamma_0)|\delta\phi|^2}{2} + \frac{|\nabla_\perp \delta A_\parallel|^2}{8\pi}\right], \quad P = \int d\mathbf{r}\delta A_\parallel \frac{1-\Gamma_0}{\rho_i^2}\delta\phi \tag{2.3}$$

Wave Kinetic Equation

- Dynamic equation (after perturbative treatments)

$$i\partial_t c_k = \int V_{12}^k c_1 c_2 e^{i\omega_{12}^k t} \delta_{12}^k d\mathbf{k}_1 d\mathbf{k}_2 \tag{2.4}$$

  - wave-action amplitude $c_k = \sqrt{\sigma_k(1-\Gamma_k)/|\omega_k|}\phi_k \equiv g_k \phi_k$
  - wave-action density $n_k = E_k/|\omega_k| = \sigma_k(1-\Gamma_k)|\phi_k|^2/|\omega_k|$
  - interaction matrix elements $V_{12}^k = \frac{\Lambda_{12}^k}{4|\omega_k|}\frac{\sigma_k\sigma_1\sigma_2}{g_k g_1 g_2}\left(\frac{k_{1\parallel}}{\omega_1} - \frac{k_{2\parallel}}{\omega_2}\right)\left(b_k\frac{k_\parallel}{\omega_k} + b_1\frac{k_{1\parallel}}{\omega_1} + b_2\frac{k_{2\parallel}}{\omega_2}\right)$

- Hierarchy of moment equations closed by random phase approximation (quasi-normal approximation) $\Longrightarrow$ wave kinetic equation (WKE) for $n_k = |c_k|^2$ [V.E. Zakharov et al. 1992]

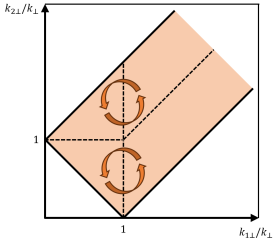
$$\partial_t n_k = \int d\mathbf{k}_1 d\mathbf{k}_2 (\mathcal{R}_{12}^k - \mathcal{R}_{2k}^1 - \mathcal{R}_{k1}^2) \tag{2.5}$$

  - integrand $\mathcal{R}_{12}^k = 4\pi|V_{12}^k|^2(n_1 n_2 - n_k n_1 - n_k n_2)\delta(\omega_{12}^k)\delta_{12}^k$
  - resonance conditions $\delta(\omega_{12}^k) = \delta(\omega_k - \omega_1 - \omega_2),\ \delta_{12}^k = \delta(\mathbf{k} - \mathbf{k}_1 - \mathbf{k}_2)$
  - wave energy $E_k = \omega_k n_k$ and wave parallel-momentum $P_k = k_\parallel n_k$ are conserved

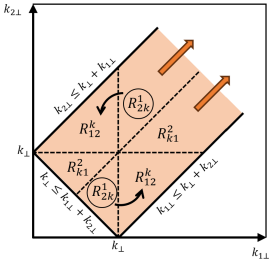
Zakharov Transformation

- Zakharov transformation (ZT) is a conformal transformation that swaps the integral intervals of WKE [V.E.Zakharov et al. 1992]
  - scale-invariant anisotropic system

$$\begin{cases} k_{1\perp} \longrightarrow k_\perp^2/k_{1\perp} \\ k_{2\perp} \longrightarrow k_\perp k_{2\perp}/k_{1\perp} \end{cases} \qquad \begin{cases} k_{1\parallel} \longrightarrow k_\parallel^2/k_{1\parallel} \\ k_{2\parallel} \longrightarrow k_\parallel k_{2\parallel}/k_{1\parallel} \end{cases} \tag{2.6}$$



ZT in $k_\perp$-space



$\mathcal{R}^1_{2k}$ is mapped onto $\mathcal{R}^k_{12}$

Kolmogorov-Zakharov Spectra

- Stationary solutions in the form $n_k \propto k_\parallel^{\nu_\parallel} k_\perp^{\nu_\perp}$

$$0 = \partial_t n_k = \int d\mathbf{k}_1 d\mathbf{k}_2 \mathcal{R}_{12}^k \left[1 - \left(\frac{k_{1\parallel}}{k_\parallel} \right)^{\mu_\parallel} \left(\frac{k_{1\perp}}{k_\perp} \right)^{\mu_\perp} - \left(\frac{k_{2\parallel}}{k_\parallel} \right)^{\mu_\parallel} \left(\frac{k_{2\perp}}{k_\perp} \right)^{\mu_\perp} \right] \tag{2.7}$$

  - $(\mu_\parallel, \mu_\perp) = (\alpha_\parallel - 2\nu_\parallel - 2\beta_\parallel - 2, \alpha_\perp - 2\nu_\perp - 2\beta_\perp - 4)$
  - $(\alpha_\parallel, \alpha_\perp)$ and $(\beta_\parallel, \beta_\perp)$ are homogeneity degrees of ω_k and V_{12}^k respectively

- Energy cascade solutions: $\delta(\omega_k - \omega_1 - \omega_2) \propto \delta \left[1 - \left(\frac{k_{1\parallel}}{k_\parallel} \right)^{\alpha_\parallel} \left(\frac{k_{1\perp}}{k_\perp} \right)^{\alpha_\perp} - \left(\frac{k_{2\parallel}}{k_\parallel} \right)^{\alpha_\parallel} \left(\frac{k_{2\perp}}{k_\perp} \right)^{\alpha_\perp} \right]$

$$\begin{cases} \mu_\parallel = \alpha_\parallel \\ \mu_\perp = \alpha_\perp \end{cases} \implies n_k \propto k_\parallel^{-\beta_\parallel - 1} k_\perp^{-\beta_\perp - 2} \tag{2.8}$$

- Parallel-momentum cascade solutions: $\delta(k_\parallel - k_{1\parallel} - k_{2\parallel}) \propto \delta \left(1 - \frac{k_{1\parallel}}{k_\parallel} - \frac{k_{2\parallel}}{k_\parallel} \right)$

$$\begin{cases} \mu_\parallel = 1 \\ \mu_\perp = 0 \end{cases} \implies n_k \propto k_\parallel^{(\alpha_\parallel - 2\beta_\parallel - 3)/2} k_\perp^{(\alpha_\perp - 2\beta_\perp - 4)/2} \tag{2.9}$$

- Parallel-momentum cascade solutions exist only in the co-propagating case

Cascade Direction

- Energy and parallel-momentum conservations

$$\partial_t E(\mathbf{k}) = -\frac{\partial \Pi_\parallel}{\partial k_\parallel} - \frac{1}{k_\perp}\frac{\partial (k_\perp \Pi_\perp)}{\partial k_\perp}, \quad \partial_t P(\mathbf{k}) = -\frac{\partial R_\parallel}{\partial k_\parallel} - \frac{1}{k_\perp}\frac{\partial R_\perp}{\partial k_\perp} \tag{2.10}$$

- Perpendicular energy flux and parallel energy flux

$$\Pi_\perp = -\frac{2\pi A^2 C_V^2}{k_\parallel k_\perp}\frac{\partial I_\xi(\nu_\parallel,\nu_\perp)}{\partial \nu_\perp}\bigg|_{\substack{\nu_\parallel=-\beta_\parallel-1\\ \nu_\perp=-\beta_\perp-2}}, \quad \Pi_\parallel = -\frac{2\pi A^2 C_V^2}{k_\perp^2}\frac{\partial I_\xi(\nu_\parallel,\nu_\perp)}{\partial \nu_\parallel}\bigg|_{\substack{\nu_\parallel=-\beta_\parallel-1\\ \nu_\perp=-\beta_\perp-2}} \tag{2.11}$$

$$R_\perp = -\frac{2\pi A^2 C_V^2}{C_\omega k_\parallel k_\perp}\frac{\partial I_\xi(\nu_\parallel,\nu_\perp)}{\partial \nu_\perp}\bigg|_{\substack{\nu_\parallel=(\alpha_\parallel-2\beta_\parallel-3)/2\\ \nu_\perp=(\alpha_\perp-2\beta_\perp-4)/2}}, \quad R_\parallel = -\frac{2\pi A^2 C_V^2}{C_\omega k_\perp^2}\frac{\partial I_\xi(\nu_\parallel,\nu_\perp)}{\partial \nu_\parallel}\bigg|_{\substack{\nu_\parallel=(\alpha_\parallel-2\beta_\parallel-3)/2\\ \nu_\perp=(\alpha_\perp-2\beta_\perp-4)/2}} \tag{2.12}$$

- We focus only on cascade directions in $k_\perp$-space according to perpendicular fluxes $\Pi_\perp$ and $R_\perp$

$$\frac{\Pi_\parallel}{\Pi_\perp} = \frac{k_\parallel}{k_\perp}\frac{\partial I_\xi/\partial \nu_\parallel}{\partial I_\xi/\partial \nu_\perp}\bigg|_{\substack{\nu_\parallel=-\beta_\parallel-1\\ \nu_\perp=-\beta_\perp-2}} \ll 1, \quad \frac{R_\parallel}{R_\perp} = \frac{k_\parallel}{k_\perp}\frac{\partial I_\xi/\partial \nu_\parallel}{\partial I_\xi/\partial \nu_\perp}\bigg|_{\substack{\nu_\parallel=(\alpha_\parallel-2\beta_\parallel-3)/2\\ \nu_\perp=(\alpha_\perp-2\beta_\perp-4)/2}} \ll 1 \tag{2.13}$$

KZ Spectra of KAW Turbulence

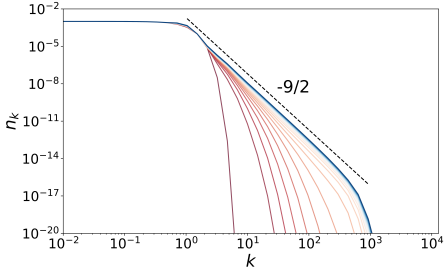
- There are three scale-invariant cases of KAW turbulence
  - $k_\perp \rho_i \gg 1 \Longrightarrow \omega_k \propto k_\parallel k_\perp, \quad |V_{12}^k| \propto \sin\theta_k \left(\frac{k_{1\parallel} k_{1\perp} k_{2\parallel} k_{2\perp}}{k_\parallel k_\perp}\right)^{1/2} |s_1 k_{2\perp} - s_2 k_{1\perp}||s_k k_\perp + s_1 k_{1\perp} + s_2 k_{2\perp}|$
  - $k_\perp \rho_i \ll 1$ (co-prop) $\Longrightarrow \omega_k - k_\parallel v_A \propto k_\parallel k_\perp^2, \quad |V_{12}^k| \propto \frac{\sin\theta_k}{k_\perp}\left(\frac{k_{1\parallel} k_{2\parallel}}{k_\parallel}\right)^{1/2} |k_{1\perp}^2 - k_{2\perp}^2|(k_\perp^2 + k_{1\perp}^2 + k_{2\perp}^2)$
  - $k_\perp \rho_i \ll 1$ (counter-prop) $\Longrightarrow \omega_k \propto k_\parallel, \quad |V_{12}^k| \propto \frac{\sin\theta_k}{k_\perp}\left(\frac{k_{1\parallel} k_{2\parallel}}{k_\parallel}\right)^{1/2} |k_\perp^2 + k_{1\perp}^2 - k_{2\perp}^2|$
- We summarize all KZ spectra and their cascade directions

regime	$(\alpha_\parallel, \alpha_\perp)$	$(\beta_\parallel, \beta_\perp)$	invariant	$(\nu_\parallel, \nu_\perp)$	direction
$k_\perp \rho_i \gg 1$	(1,1)	(1/2,5/2)	energy	(-3/2,-9/2)	direct
			momentum	(-3/2,-4)	inverse
$k_\perp \rho_i \ll 1$ (co-prop)	(1,2)	(1/2,3)	energy	(-3/2,-5)	direct
			momentum	(-3/2,-4)	inverse
$k_\perp \rho_i \ll 1$ (counter-prop)	(1,0)	(1/2,1)	energy	(-3/2,-3)	direct

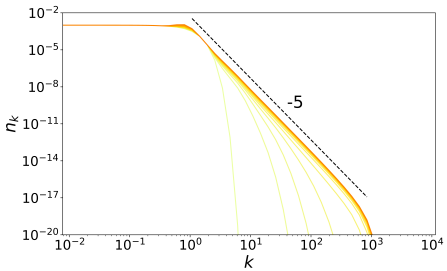
- Note that $n_k \propto k_\parallel^{-3/2} k_\perp^{-3}$ and $n_k \propto k_\parallel^{-3/2} k_\perp^{-9/2}$ for counter-propagating (balanced) case were verified by [S.Boldyrev & J.C.Perez PRL 2009; V.David et al. PRL 2024]

Numerical Verification

- Assuming $n_k \propto k_\parallel^{-3/2} n(k_\perp)$, we concentrate on the $k_\perp$-cascade for co-propagating case.
- The direct-energy-cascade spectra for co-propagating case are verified.



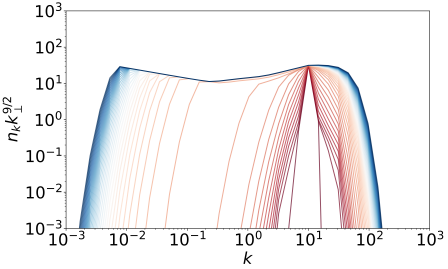
The formation of $k_\perp^{-9/2}$ stationary spectrum for the $k_\perp \rho_i \gg 1$ (co-prop) case



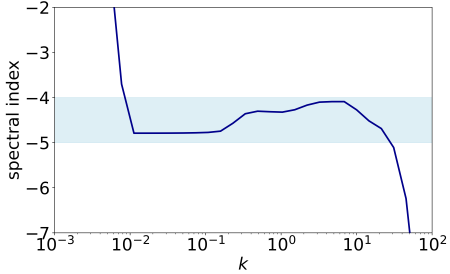
The formation of $k_\perp^{-5}$ stationary spectrum for the $k_\perp \rho_i \ll 1$ (co-prop) case

Numerical Verification

- An example of co-propagating case with complete FILR effect:
  - an inverse cascade observed with a small-scale injection, with spectral index lying between -4 and -5.
  - a steeper spectrum at large scales implies the potential existence of nonlocal transfer.



Compensated stationary spectrum



Spectral index of the stationary spectrum

Conclusion

- Wave-kinetic description of weak KAW turbulence based on GK theory proposed.
- Stationary spectra analytically obtained using Zakharov transformation in both $k_\perp \rho_i \ll 1$ and $k_\perp \rho_i \gg 1$ limits, for both counter-propagating and co-propagating cases:

regime	$(\alpha_\parallel, \alpha_\perp)$	$(\beta_\parallel, \beta_\perp)$	invariant	$(\nu_\parallel, \nu_\perp)$	direction
$k_\perp \rho_i \gg 1$	(1,1)	(1/2,5/2)	energy	(-3/2,-9/2)	direct
			momentum	(-3/2,-4)	inverse
$k_\perp \rho_i \ll 1$ (co-prop)	(1,2)	(1/2,3)	energy	(-3/2,-5)	direct
			momentum	(-3/2,-4)	inverse
$k_\perp \rho_i \ll 1$ (counter-prop)	(1,0)	(1/2,1)	energy	(-3/2,-3)	direct

- Direct-energy-cascade spectra for co-propagating case numerically verified.
- Inverse cascade due to parallel-momentum conservation and its locality should be investigated in detail (future work).

Thank you for your attention!

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Submitted to ApJ